Recommendations — ulms-highgrade-pulm-mets-er-pr-neg-qqvt

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-08-21

Recommendations — ulms-highgrade-pulm-mets-er-pr-neg-qqvt

Profile snapshot

- **Primary site:** uterus (myometrium)
- **Histology:** high-grade uterine leiomyosarcoma (uLMS), spindle-cell morphology; diagnosis confirmed on both the pulmonary biopsy and the re-reviewed uterine specimen by a reference pathology laboratory (August 2026)
- **Stage:** metastatic (stage IV): bilateral pulmonary metastases. Radiologic read by the consulting sarcoma oncologist: lesions unusually infiltrative and possibly inflammatory in appearance. Primary resected September 2024 (signed out benign at the time); metastatic diagnosis July-August 2026
- **ECOG:** 1
- **Age band:** 30-39
- **Sex:** female
- **Biomarkers:** ER (estrogen receptor) <1% positive (decision-relevant negative); PR (progesterone receptor) <1% positive (decision-relevant negative); TP53 not tested (no comprehensive genomic profiling reported) (ngs_pending); RB1 not tested (no comprehensive genomic profiling reported) (ngs_pending); ATRX not tested (no comprehensive genomic profiling reported) (ngs_pending); MED12 not tested (no comprehensive genomic profiling reported) (ngs_pending); Homologous recombination status incl. BRCA2 not tested (no comprehensive genomic profiling reported) (ngs_pending); NTRK fusion not tested (no comprehensive genomic profiling reported) (ngs_pending); TMB (tumor mutational burden) not tested (no comprehensive genomic profiling reported) (ngs_pending); MSI / MMR not tested (ngs_pending); PD-L1 not tested (ihc_pending)
- **Efficacy/toxicity weight:** 0.7

22 ranked therapeutic options across 10 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Ulms Histology interventions

1. Doxorubicin + trabectedin induction, then trabectedin maintenance (LMS-04 regimen), first line

Likelihood of effect: High: adjusted HR 0.41 for PFS (BICR primary) and 0.65 for OS in randomized first-line LMS (pmid:35835135, pmid:39231341); maintenance contribution unmeasured.

Toxicity burden: High (G3-4 neutropenia 80%, thrombocytopenia 47%, febrile neutropenia 28%; maintenance-phase transaminitis and rare rhabdomyolysis carried from the single-agent trabectedin dossier)

Counter-productive MoA:Low (Preclinical pairing was additive at best; untested maintenance continuation could accrue toxicity without measured contribution.)

Overall:The planned regimen and the best-evidenced option coincide; gated on cardiac clearance and an off-label authorization, with the maintenance phase the least-evidenced part.

Key references: PMID 35835135 · PMID 39231341 · NCT02997358

2. Abemaciclib + gemcitabine (NCT06498648, first-line LMS cohort) — considered and declined

Likelihood of effect: Not assessable: no LMS efficacy data, and the concurrent schedule contradicts the sequential-priming evidence (pmid:22751436, pmid:28619757) that motivated the pairing.

Toxicity burden: Moderate (no combination safety data in LMS; overlapping myelosuppression from both agents expected)

Counter-productive MoA:Moderate (Concurrent CDK4/6 inhibition shields RB-proficient cells from gemcitabine, the opposite of the sequential priming that worked in LMS xenografts.)

Overall:Declined by its own sponsor persona: the schedule argues against its own mechanism, and the chemo-naive slot it would spend is the asset that buys the 33-month arm.

Key references: NCT06498648 · PMID 22751436 · PMID 28619757

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Ivonescimab (PD-1 x VEGF bispecific; NCT07516925, phase 2 LMS, 1-3 prior lines) — _Thin evidence_

Likelihood of effect: Unknown: no LMS efficacy data for the construct; recruiting with ORR primary.

Toxicity burden: Moderate (PD-1 and VEGF class toxicities: immune-related AEs, hypertension, proteinuria)

Counter-productive MoA:Low (No mechanism objection raised; checkpoint activity in unselected LMS has historically been modest.)

Overall:A recruiting LMS-named phase 2 for the post-LMS-04 conversation, with nothing published to weigh yet.

Key references: NCT07516925

All-trans retinoic acid + cemiplimab (NCT06528769, phase 2 LMS after standard chemotherapy) — _Thin evidence_

Likelihood of effect: Unknown: no clinical efficacy for the combination in LMS; mechanism rests on preclinical MDSC work (pmid:12907617).

Toxicity burden: Moderate (checkpoint-class immune-related AEs plus ATRA headache, dryness, hypertriglyceridemia)

Counter-productive MoA:Low (No mechanism objection raised; the MDSC-differentiation premise is untested in sarcoma patients.)

Overall:A mechanism-first checkpoint combination recruiting in her disease, with the evidence still

entirely preclinical.

Key references: NCT06528769 · PMID 12907617

Zanzalintinib (XL092 multi-kinase TKI; NCT06571734, phase 2 LMS cohort, >=2 prior lines) — _Thin evidence_

Likelihood of effect: Unknown in LMS: no published data; the approved class comparator (pazopanib) sets the bar on the Standard-of-care table.

Toxicity burden: Moderate (VEGFR-TKI class: hypertension, diarrhea, fatigue, hand-foot syndrome)

Counter-productive MoA:Low (No mechanism objection; class wound-healing impairment would matter if metastasectomy ever became real.)

Overall:A third-line TKI trial door whose whole case is beating an approved drug that already exists for the same slot.

Key references: NCT06571734

Lurbinectedin + doxorubicin (SaLuDo phase 3, NCT06088290, first-line LMS) — _Not enrollable_

Likelihood of effect: Not enrollable: closed to accrual, no results posted; informational as the nearest LMS-04 replication read.

Toxicity burden: Moderate (expected to mirror anthracycline-plus-minor-groove-binder myelosuppression; no published rates)

Counter-productive MoA:Low (No mechanism objection; the open question is replication, not antagonism.)

Overall:The trial whose readout will either replicate the rank-2 effect in class or complicate it; watch, cannot join.

Key references: NCT06088290

Anlotinib / catequentinib (multi-kinase TKI; APROMISS NCT03016819, LMS indications closed) — _Not enrollable_

Likelihood of effect: Not enrollable: LMS indications closed, no US marketing authorization.

Toxicity burden: Moderate (VEGFR-TKI class: hypertension, hand-foot syndrome, fatigue)

Counter-productive MoA:Low (No mechanism objection; access is the whole problem.)

Overall:A closed trial door for a drug with no US route; the class question is answered by an approved alternative.

Key references: NCT03016819

Pegylated liposomal doxorubicin maintenance (MELODY NCT06981637, Taiwan) — _Not enrollable_

Likelihood of effect: Not enrollable (geography): Taiwan-only; informational for the maintenance question the rank-2 regimen cannot answer.

Toxicity burden: Moderate (PLD class: hand-foot syndrome, mucositis, cumulative cardiac exposure)

Counter-productive MoA:Low (No mechanism objection; the relevance is the maintenance design question, not the drug.)

Overall:The only randomized read anywhere on maintenance after anthracycline control, running an ocean away.

Key references: NCT06981637

Evidence in detail

Doxorubicin + trabectedin induction, then trabectedin maintenance (LMS-04 regimen), first line

Pautier et al. *Lancet Oncol* 2022

- **Disease context:metastatic or unresectable uterine or soft-tissue leiomyosarcoma, chemotherapy-naive** (1L) — ECOG 0-1 required; 67 of 150 patients (45%) had uterine LMS; randomization stratified by uterine vs soft-tissue origin; surgery for residual disease allowed after 6 cycles in both arms
- **n:** 150
- **Toxicities:neutropenia** (grade >=3): 80% (n=74); **thrombocytopenia** (grade >=3): 47% (n=74); **anemia** (grade >=3): 31% (n=74); **febrile neutropenia** (grade >=3): 28% (n=74); **serious adverse event** (grade any): 20% (n=74); **treatment-related death (cardiac failure)** (grade 5): 0% (n=74)
- **Efficacy:PFS** 12.2 months (95% CI 10.1-15.6); **PFS** 6.2 months (95% CI 4.1-7.1); **HR_PFS** 0.41 (95% CI 0.29-0.58)

Pautier et al. *N Engl J Med* 2024

- **Disease context:metastatic or unresectable uterine or soft-tissue leiomyosarcoma, chemotherapy-naive** (1L) — same 150-patient ITT population as the 2022 primary report; analyses adjusted for uterine vs soft-tissue origin and disease stage
- **n:** 150
- **Toxicities:** Abstract reports only that adverse events and dose reductions were more frequent with doxorubicin-trabectedin; per-term combination-phase rates are on the 2022 primary-report row.
- **Efficacy:OS** 33 months (95% CI 26-48); **OS** 24 months (95% CI 19-31); **HR_OS** 0.65 (95% CI 0.44-0.95); **PFS** 12 months (95% CI 10-16); **HR_PFS** 0.37 (95% CI 0.26-0.53)

Hrd interventions

1. Olaparib + temozolomide, queued for first progression (NCI protocol 10250 regimen), HRR/HRD-conditional *(conditional on hrd positive)*

Likelihood of effect: Moderate if HRD confirms: ORR 27%, mPFS 6.9 months on n=22 single-arm (pmid:37467452); exploratory RAD51 split 11.2 vs 5.4 months, assay not orderable.

Toxicity burden: High (G3-4 neutropenia 75% per abstract vs 59% per full-text table, thrombocytopenia 32%, anemia 32%; no treatment-related deaths)

Counter-productive MoA:Low (In HR-proficient tumors the synergy premise collapses toward temozolomide alone; marrow stacking after induction is the practical brake.)

Overall:The strongest non-cytotoxic signal in uLMS, opened or mostly closed by an HRR panel nobody has sent; n=22 and an unorderable stratifier bound the confidence.

Key references: PMID 37467452 · PMID 37143137 · NCT05432791

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Trabectedin + olaparib (preclinical combination; PARP1-expression-dependent synergy) — _Thin evidence_

Likelihood of effect: Not assessable: preclinical only (pmid:28454547); the proposed predictor, PARP1 expression, has never been measured in her.

Toxicity burden: Moderate (would stack trabectedin hepatotoxicity on PARP-inhibitor cytopenias; no clinical characterization)

Counter-productive MoA:Low (No mechanism objection; the combination is untested rather than contradicted.)

Overall:A preclinical synergy note connecting the rank-2 and rank-3 mechanisms, with no trial to put it in.

Key references: PMID 28454547

Evidence in detail

Olaparib + temozolomide, queued for first progression (NCI protocol 10250 regimen), HRR/HRD-conditional

Ingham et al. *J Clin Oncol* 2023

- **Disease context:advanced uterine leiomyosarcoma after >=1 prior line** (2L+) — 22 evaluable; median age 55; 59% had >=3 prior lines; no HRD entry gate — all comers with paired biopsies (WES/RNAseq + RAD51 foci assay)
- **n:** 22
- **Toxicities:neutropenia** (grade >=3): 75% (n=22); **thrombocytopenia** (grade >=3): 32% (n=22); **anemia** (grade >=3): 32% (n=22); **discontinuation for toxicity** (grade any): 14% (n=22); **olaparib dose reduction** (grade any): 36% (n=22)
- **Efficacy:ORR** 27%; **PFS** 6.9 months; **DoR** 12.0 months; **PFS** 11.2 months; **biomarker** 31%

B7H3 interventions

1. B7-H3 ADC approach (MGC026, NCT06242470, named sarcoma cohort; approach-level rank consolidating GSK5764227/HS-20093), stain-gated referral at second line or later

Likelihood of effect: Unknown in LMS: no clinical efficacy for any B7-H3 agent; 97% sarcoma positivity with 69% high (pmid:39478506) makes the gating stain worth one slide.

Toxicity burden: Moderate (no published MGC026 grade profile; predecessor duocarmycin ADC halted on safety findings; ADC-class cytopenias and infusion reactions expected)

Counter-productive MoA:Moderate (Positivity is not dependency; activity fell away below an antigen-density threshold, so low expressors likely gain little.)

Overall:The most reachable experimental door if the stain is strongly positive, on a target class whose predecessor died on safety and which has no LMS response data.

Key references: NCT06242470 · PMID 39478506 · PMID 37775116

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

GSK5764227 / HS-20093 (B7-H3 ADC; EMBOLD NCT06551142, ARTEMIS-002 NCT05830123) — _Consolidated_

Likelihood of effect: See rank 4: the approach is ranked once; no LMS response data for any B7-H3 agent.

Toxicity burden: Moderate (topoisomerase-payload ADC class profile; no LMS-specific data)

Counter-productive MoA:Moderate (Same antigen-density dependency as the ranked B7-H3 approach.)

Overall:The backup sponsor for the rank-4 approach, reachable mainly outside the US.

Key references: NCT06551142 · NCT05830123

Vobramitamab duocarmazine (B7-H3 duocarmycin ADC; NCT05551117, terminated) — _Unavailable_

Likelihood of effect: Not obtainable: programme terminated on safety findings; its bystander data (pmid:37775116) are why the class stays interesting.

Toxicity burden: High (programme halted after safety findings in the TAMARACK context)

Counter-productive MoA:Moderate (The payload liability that ended this programme may be a class property; the successor has not yet shown otherwise.)

Overall:The corpse in the B7-H3 class: its preclinical data motivate rank 4 and its termination is rank 4's biggest caveat.

Key references: NCT05551117 · PMID 37775116

Fap interventions

1. FAP-targeted radioligand approach ([225Ac]RTX-2358, NCT07156565, FAPi-PET-gated; approach-level rank consolidating [177Lu]Lu-FAP-2286 / LuMIERE), screen at first progression

Likelihood of effect: Unknown: no human sarcoma efficacy for this agent; tumour-cell FAP >=2 intensity in 50.7% of sarcomas (pmid:37333052) and PDX regressions support the screen.

Toxicity burden: Moderate (no human grade profile; on-target marrow-stroma concern from FAP CAR-T mouse work; radioligand-class cytopenias expected)

Counter-productive MoA:Moderate (FAP sits on marrow stroma; the crossfire that beats tumour heterogeneity reaches it too, so dosimetry rather than IHC decides.)

Overall:A forward-running gate at zero tissue cost: a FAPi PET either finds the target or closes the branch, with marrow-stroma dosimetry the open safety question.

Key references: NCT07156565 · PMID 37333052 · PMID 23712432

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

[177Lu]Lu-FAP-2286 (beta-emitting FAP radioligand; LuMIERE NCT04939610) — _Consolidated_

Likelihood of effect: See rank 5: the approach is ranked once; sarcoma PDX regressions (pmid:35608703) belong to this compound.

Toxicity burden: Moderate (no human sarcoma grade profile; radioligand-class cytopenias expected)

Counter-productive MoA:Moderate (Same marrow-stroma FAP exposure question as the ranked alpha emitter, at lower per-decay energy.)

Overall:The beta-emitter form of the rank-5 approach, and possibly the gentler first pass if uptake confirms.

Key references: NCT04939610 · PMID 35608703

Rb1 Loss interventions

1. REC-617 (selective CDK7 inhibitor) after RB1-null IHC, second line (NCT07633756, single site)

Likelihood of effect: Unknown: zero CDK7 data in LMS; mechanism borrowed from RB1-null SCLC with a different compound (pmid:31883968). Phase 1b, safety primary.

Toxicity burden: Moderate (no published REC-617 grade profile; CDK7-class cytopenias and GI toxicity expected; dose escalation design)

Counter-productive MoA:Moderate (The gating stain routes a near-universally RB1-null histology into the arm with no LMS data; mechanism is cross-lineage borrowed.)

Overall:A one-slide stain opens a first-in-class door, but the gate runs backwards in this histology and the arm it opens holds no LMS data.

Key references: NCT07633756 · PMID 31883968

Prame Hla interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

IMA203 / IMA203CD8 (PRAME TCR-T, HLA-A*02:01-restricted; NCT03686124) — _Thin evidence_

Likelihood of effect: Unknown: both gates (HLA-A*02:01 and PRAME expression) unmeasured; LMS PRAME prior runs low.

Toxicity burden: High (TCR-T class: cytokine release syndrome, lymphodepletion cytopenias)

Counter-productive MoA:Low (Antigen-loss escape and low LMS PRAME density are the class risks; no persona assessed it.)

Overall:A cell-therapy door that costs a blood draw and two slides to price, with a low prior in this histology.

Key references: NCT03686124

Brenetafusp (PRAME x CD3 ImmTAC; NCT07686367, synovial sarcoma / MRCLS only) — _Not enrollable_

Likelihood of effect: Not enrollable: the sarcoma protocol excludes LMS; PRAME and HLA both unmeasured besides.

Toxicity burden: Moderate (ImmTAC-class cytokine release and skin toxicity expected)

Counter-productive MoA:Low (Class risks (antigen density, T-cell fitness) unexamined in LMS; no persona assessed it.)

Overall:A PRAME door built for a different sarcoma; the workup keeps the thread without booking anything.

Key references: NCT07686367

Her2 interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Trastuzumab deruxtecan (T-DXd; DESTINY-PanTumor02 basket, NCT04482309) — _Thin evidence_

Likelihood of effect: Low prior: HER2 unmeasured, no sarcoma cohort in DESTINY-PanTumor02 (pmid:37870536); IHC 3+ would change the table it belongs on.

Toxicity burden: Moderate (interstitial lung disease/pneumonitis, a named risk against bilateral infiltrative lung disease; cytopenias, nausea)

Counter-productive MoA:Moderate (Drug-related pneumonitis would be unreadable against her infiltrative lung disease and altitude-shifted oximetry.)

Overall:An unmeasured gate with a low prior, and a pneumonitis liability her lungs are poorly placed to absorb.

Key references: PMID 37870536 · NCT04482309

Atrx Alt interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

ATR inhibition for ATRX-loss / ALT-positive tumors (mechanism hypothesis) — _Thin evidence_

Likelihood of effect: Low: the founding sensitivity claim (pmid:25593184) failed replication twice (pmid:27602331, pmid:33344901).

Toxicity burden: Low (no clinical vehicle in this dossier to attach a grade profile to)

Counter-productive MoA:Moderate (The founding synthetic-lethality claim did not replicate, so the mechanism may simply not be there.)

Overall:A twice-failed replication wearing a plausible mechanism; the workup reports ATRX for prognosis, not for this.

Key references: PMID 25593184 · PMID 27602331 · PMID 33344901

Magea4 Hla interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Afamitresgene autoleucel (MAGE-A4 TCR-T, approved in synovial sarcoma / MRCLS; SPEARHEAD-1) — _Not enrollable_

Likelihood of effect: Not applicable to LMS: licensed indication excludes her histology; both gates unmeasured and the antigen prior is low.

Toxicity burden: High (TCR-T class: cytokine release syndrome in most patients in SPEARHEAD-1, cytopenias)

Counter-productive MoA:Low (Low MAGE-A4 antigen density in LMS is the class problem; no persona assessed it.)

Overall:An approved cell therapy pointed at a different sarcoma; for her it is two stains and a typing result away from even being a question.

Key references: PMID 38554725 · PMID 36624315

Pulmonary Metastases Pattern interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Pulmonary suffusion + metastasectomy (NCT03965234, phase 1/2 feasibility) — _Not enrollable_

Likelihood of effect: Not currently a candidate: bilateral infiltrative disease is what the surgical series screened out; feasibility endpoint only.

Toxicity burden: Moderate (thoracic surgical morbidity; no efficacy-grade data)

Counter-productive MoA:Low (No mechanism objection; anatomy and staging are the gates.)

Overall:A surgical-adjunct trial for an anatomy she does not currently have, behind a staging question the workup settles.

Key references: NCT03965234